Date: / /

3.) Working Principle and Depth Calculation.

The stereo-camera uses a technique called triangulation to calculate depth. It identifies the same point in both images and measures the difference in its horizontal positions, called disparity.

Depth is calculated via:

$$Z = \frac{f \times B}{d}$$

where,
Z = Distance/depth of the object.
f = focal length of camera
B = Distance between 2 cameras. (baseline)
d = Disparity between the 2 images.

Conclusion:

Stereo cameras provide with 3D vision & the ability to estimate real-world distances. They can detect obstacles, identify ditches and help robots navigate their surroundings. When combined with AI, they become useful tools for intelligent perception & autonomous navigation.

Date: / /

viewpoints & compares them to identify obstacles. It calculates the depth of objects & create a 3-D representation of the surroundings. AI algorithms can classify detected objects and distinguish obstacles from the background. The robot can then determine their distance and plan a suitable path to avoid collisions.

4.) Can a stereo camera detect & measure a ditch?

A stereo-camera can detect a ditch / depression in front of a robot by analysing changes in ground depth. It compares the expected ground surface with the actual depth measurements to identify sudden drops or missing ground regions. AI can classify these regions as potential ditches and estimate their depth and width.

5.) Limitations / flaws of stereo-cameras.

Stereo cameras may produce inaccurate depth measurements in poor lighting or on textureless and reflective surfaces. Their accuracy generally decrease as the distance from the camera increases. Object hidden behind other objects may also be difficult to detect. So, combining stereo-cameras with sensors such as LiDAR can improve reliability.

papergrid

Date: / /

the

& predict robot's movements.

3) Inverse kinematics.

→ Inverse kinematics determines the joint angles required to reach a desired position and orientation.

→ It works backwards from the target position of the end effector.

→ It enables the robot to reach objects & perform specific tasks accurately.

→ AI can use inverse kinematics to plan & optimize the robot's movements.

Freedom & kinematics.

Name: Alister D'Souza. Date: 30-10-2024.

1) Degrees of freedom (DOF):

→ Degrees of freedom refers to the number of independent movements a robotic manipulator can perform.

→ Exact joint contributes to the robot's overall mobility.

→ A gripper may add an additional degree of freedom through its opening and closing mechanism.

~~4.)~~ Forward • In AI, D·O·F helps determine the robot's movement capabilities and workspace.

2.) Forward kinematics.

→ Forward kinematics determines the position and orientation of a robot's end effector.

→ It uses known joint angles and the dimensions of the robot's links.

→ It helps calculate where the gripper will be located after joint movements.

→ AI can use forward kinematics to monito

STEREOCAMERA

Name: Alister D. Dsouza

Date: 30-10-2026.

1) Introduction:

A stereo camera is a vision system that uses 2 cameras placed from each other at a fixed distance apart to capture images of the same surroundings. It works similarly to human binocular vision. By comparing the 2 images, it computes and estimates the distance and depth of objects. Stereo-cameras are widely used in robotics, autonomous vehicles & AI-based obstacle detection.

2) Can a Stereo-camera measure actual depth?

A stereo camera can measure the actual depth of objects in real-world units, such as m or cm. It computes the depth by comparing the difference in the position of an object in the 2 images, known as disparity. Using the camera calibration and the distance between the cameras, it estimates the objects distance.

3) How does a stereocamera detect hurdles?

A stereo camera captures images from 2 different